ICE-4107: Microwave Engineering

Lecture 4

Electromagnetic Plane Waves

Course Teacher

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The Helmholtz Equation

- In a source-free, linear, isotropic, homogeneous region, Maxwell's curl equations in phasor form are

$$\nabla \times \bar{E} = -j\omega\mu \bar{H}, \tag{1.41a}$$

$$\nabla \times \bar{H} = j\omega\epsilon \bar{E}, \tag{1.41b}$$

and constitute two equations for the two unknowns, $\bar{E}$ and $\bar{H}$. As such, they can be solved for either $\bar{E}$ or $\bar{H}$. Taking the curl of (1.41a) and using (1.41b) gives

$$\nabla \times \nabla \times \bar{E} = -j\omega\mu \nabla \times \bar{H} = \omega^2 \mu\epsilon \bar{E},$$

which is an equation for $\bar{E}$. This result can be simplified through the use of vector identity (B.14), $\nabla \times \nabla \times \bar{A} = \nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A}$, which is valid for the rectangular components of an arbitrary vector $\bar{A}$. Then,

$$\nabla^2 \bar{E} + \omega^2 \mu \epsilon \bar{E} = 0, \tag{1.42}$$

since $\nabla \cdot \bar{E} = 0$ in a source-free region. Equation (1.42) is the wave equation, or Helmholtz equation, for $\bar{E}$. An identical equation for $\bar{H}$ can be derived in the same manner:

$$\nabla^2 \bar{H} + \omega^2 \mu \epsilon \bar{H} = 0. \tag{1.43}$$

A constant $k = \omega\sqrt{\mu\epsilon}$ is defined and called the wavenumber, or propagation constant, of the medium; its units are 1/m.

As a way of introducing wave behavior, we will next study the solutions to the above wave equations in their simplest forms, first for a lossless medium and then for a lossy (conducting) medium.

Activate Win

Poynting's theorem

- In electrodynamics, **Poynting's theorem** is a statement of conservation of energy for electromagnetic fields developed by British physicists John Henry Poynting. It states that in a given volume, the stored energy changes at a rate given by the work done on the charges within the volume, minus the rate at which energy leaves the volume.

Poynting's theorem

- From the standpoint of complex power in terms of the complex field vectors, the time average of any two complex vectors is equal to the real part of the product of one complex vector multiplied by the complex conjugate of the other vector. Hence the time average of the instantaneous Poynting vector in steady state is given by

$$\langle P \rangle = \langle \mathbf{E} \times \mathbf{H} \rangle = \tfrac{1}{2}\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*) \qquad (2\text{-}2\text{-}1)$$

where the notation < > stands for the average and the factor of 1/2 appears in the equation for complex power when peak values are used for the complex quantities E and H. Re represents the real part of the complex power, and the asterisk indicates the complex conjugate

Poynting's theorem

It is necessary to define a complex Poynting vector as

$$\mathbf{P} = \tfrac{1}{2}(\mathbf{E} \times \mathbf{H}^*) \tag{2-2-2}$$

Maxwell's equations in frequency domain for the electric and magnetic fields are

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} \tag{2-2-3}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + j\omega\epsilon\mathbf{E} \tag{2-2-4}$$

Dot multiplication of Eq. (2-2-3) by $\mathbf{H}^*$ and of the conjugate of Eq. (2-2-4) by $\mathbf{E}$ yields

$$(\nabla \times \mathbf{E}) \cdot \mathbf{H}^* = -j\omega\mu\mathbf{H} \cdot \mathbf{H}^* \tag{2-2-5}$$

$$(\nabla \times \mathbf{H}^*) \cdot \mathbf{E} = (\mathbf{J}^* - j\omega\epsilon\mathbf{E}^*) \cdot \mathbf{E} \tag{2-2-6}$$

Then subtraction of Eq. (2-2-5) from Eq. (2-2-6) results in

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}^*) - \mathbf{H}^* \cdot (\nabla \times \mathbf{E}) = \mathbf{E} \cdot \mathbf{J}^* - j\omega(\epsilon|E|^2 - \mu|H|^2) \tag{2-2-7}$$

where $\mathbf{E} \cdot \mathbf{E}^*$ is replaced by $|E|^2$ and $\mathbf{H} \cdot \mathbf{H}^*$ by $|H|^2$.

Poynting's theorem

The left-hand side of Eq. (2-2-7) is equal to $-\nabla \cdot (\mathbf{E} \times \mathbf{H}^*)$ by the vector identity. So

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}^*) = -\mathbf{E} \cdot \mathbf{J}^* + j\omega(\epsilon|E|^2 - \mu|H|^2) \qquad (2\text{-}2\text{-}8)$$

The electric current density includes two components—that is,

$$\mathbf{J} = \mathbf{J}_c + \mathbf{J}_0 \qquad (2\text{-}1\text{-}5)$$

where $\mathbf{J}_c = \sigma \mathbf{E}$ is called the *conduction current density*
$\mathbf{J}_0$ = the *impressed current density*, which is independent of the field

Substituting Eqs. (2-1-5) and (2-2-2) into Eq. (2-2-8), we have

$$-\tfrac{1}{2}\mathbf{E} \cdot \mathbf{J}_0^* = \tfrac{1}{2}\sigma\mathbf{E} \cdot \mathbf{E}^* + j\omega(\tfrac{1}{2}\mu\mathbf{H} \cdot \mathbf{H}^* - \tfrac{1}{2}\epsilon\mathbf{E} \cdot \mathbf{E}^*) + \nabla \cdot \mathbf{P} \qquad (2\text{-}2\text{-}9)$$

Integration of Eq. (2-2-9) over a volume and application of Gauss's theorem to the last term on the right-hand side give

$$\int_v \tfrac{1}{2}(\mathbf{E} \cdot \mathbf{J}_0^*)dv = \int_v \tfrac{1}{2}\sigma|E|^2 dv + j2\omega \int_v (w_m - w_e)dv + \oint_S \mathbf{P} \cdot d\mathbf{s} \qquad (2\text{-}2\text{-}10)$$

Poynting's theorem

where $\frac{1}{2}\sigma|E|^2 = \sigma\langle|E|^2\rangle$ is the time-average dissipated power

$\frac{1}{4}\mu\mathbf{H}\cdot\mathbf{H}^* = \frac{1}{2}\mu\langle|H|\rangle^2 = w_m$ is the time-average magnetic stored energy

$\frac{1}{4}\epsilon\mathbf{E}\cdot\mathbf{E}^* = \frac{1}{2}\epsilon\langle|E|^2\rangle = w_e$ is the time-average electric stored energy

$-\frac{1}{2}\mathbf{E}\cdot\mathbf{J}_0^*$ = the complex power impressed by the source $\mathbf{J}_0$ into the field

Equation (2-2-10) is well known as the *complex Poynting theorem* or the *Poynting theorem* in frequency domain.

Furthermore, let

$P_{in} = -\int_v \frac{1}{2}(\mathbf{E}\cdot\mathbf{J}_0^*)dv$ be the total complex power suplied by a source within a region

$\langle P_d\rangle = \int_v \frac{1}{2}\sigma|E|^2 dv$ be the time-average power dissipated as heat inside the region

Poynting's theorem

$$\langle W_m - W_e \rangle = \int_v (w_m - w_e) dv$$ be the difference between time-average magnetic and electric energies stored within the region

$$P_{tr} = \oint \mathbf{P} \cdot d\mathbf{s}$$ be the complex power transmitted from the region

The complex Poynting theorem shown in Eq. (2-2-10) can be simplified to

$$P_{in} = \langle P_d \rangle + j2\omega[\langle W_m - W_e \rangle] + P_{tr} \qquad (2\text{-}2\text{-}11)$$

This theorem states that the total complex power fed into a volume is equal to the algebraic sum of:

- Active power dissipated as heat,
- Reactive power proportional to the difference between time-average magnetic and electric energies stored in the volume,
- Complex power transmitted across the surface enclosed by the volume.

Uniform Plane Waves

- A plane wave is a wave whose phase is constant over a set of planes.
- A uniform plane wave is a wave whose magnitude and phase are both constant.
- A spherical wave in free space is a uniform plane wave as observed at a far distance.
- Its equiphase surfaces are concentric spheres, expanding as the wave travels outward from the source, and its magnitude is constant.
- Electromagnetic waves in free space are typical uniform plane waves.

Uniform Plane Waves

- The electric and magnetic fields are mutually perpendicular to each other and to the direction of propagation of the waves.

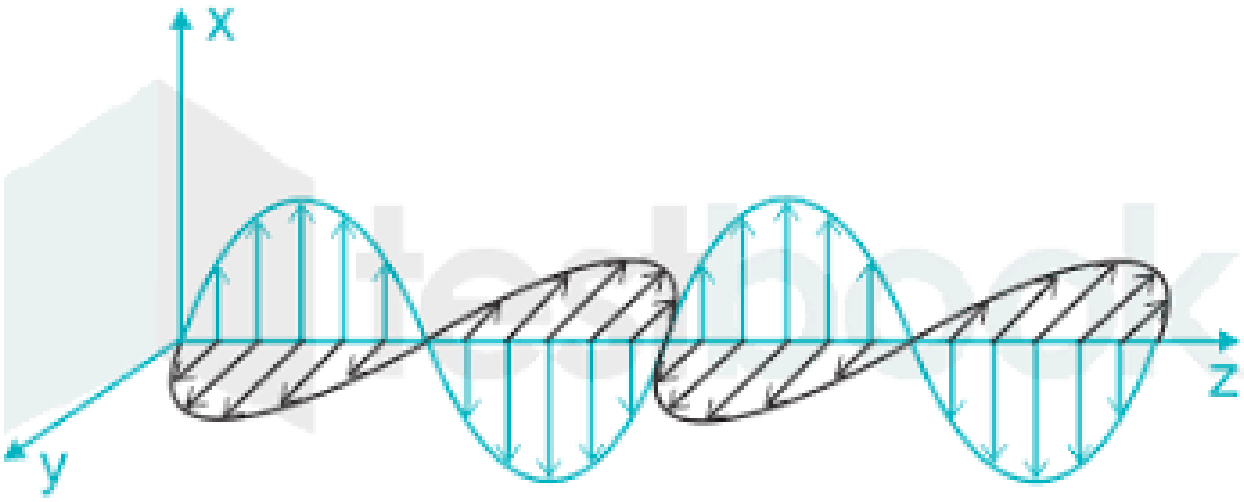


Fig: Uniform plane wave

Uniform Plane Waves

- The phases of the two fields are always in time phase and their magnitudes are always constant.
- The stored energies are equally divided between the two fields, and the energy flow is transmitted by the two fields in the direction of propagation.
- A uniform plane wave is a transverse electromagnetic wave or a TEM wave.
- A nonuniform plane wave is a wave whose amplitude (not phase) may vary within a plane normal to the direction of propagation. Consequently, the electric and magnetic fields are no longer in time phase.

Uniform Plane Waves

- Since a uniform plane wave of electric or magnetic field has no variation of intensity in a plane normal to the direction of propagation of the wave, then

$$\frac{\partial \mathbf{E}}{\partial x} = \frac{\partial \mathbf{E}}{\partial y} = 0 \quad \text{or} \quad \frac{\partial \mathbf{H}}{\partial x} = \frac{\partial \mathbf{H}}{\partial y} = 0$$

if the direction of propagation is assumed in the positive z direction.

With the preceding assumptions and lossless dielectric, that is, $\sigma = 0$, the wave equations (2-1-25) and (2-1-26) in time domain for the electric and magnetic intensities in free space for rectangular coordinates

Uniform Plane Waves

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \tag{2-1-25}$$

$$\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2} \tag{2-1-26}$$

reduce to:

$$\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2} \tag{2-3-1}$$

$$\frac{\partial^2 H_y}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 H_y}{\partial t^2} \tag{2-3-2}$$

in which the electric intensity is arbitrarily chosen in the *x* direction and the magnetic intensity in the *y* direction. With no loss in generality, the electric intensity is given by:

- Uniform Plane Waves

$$E_x = E_0 e^{j(\omega t - \beta z)} = E_0 e^{j\omega\{t-(\beta/\omega)z\}} \tag{2-3-3}$$

The magnetic intensity can be obtained by inserting Eq. (2-3-3) into the curl equation

$$\nabla \times \mathbf{E} = -j\omega\mu_0 \mathbf{H} \tag{2-3-4}$$

For the assumed conditions, the curl equation reduces to

$$\frac{\partial E_x}{\partial z} = -j\omega\mu_0 H_y \tag{2-3-5}$$

Differentiation of Eq. (2-3-3) with respect to z and substitution of the result in Eq. (2-3-5) yield

$$H_y = \sqrt{\frac{\epsilon_0}{\mu_0}}\, E_x \tag{2-3-6}$$

Uniform Plane Waves

where $\frac{\beta}{\omega} = \sqrt{\mu_0 \epsilon_0} = \frac{1}{v_p}$ is accounted for in the derivation

$v_p = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8$ m/s is phase velocity

$= c$, the velocity of light in vacuum

$\alpha = 0$

$\beta = \omega \sqrt{\mu_0 \epsilon_0}$ is phase constant

The ratio of electric to magnetic intensities is given by

$$\frac{E_x}{H_y} = \eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377\ \Omega \qquad (2\text{-}3\text{-}7)$$

It is called the *intrinsic impedance* of free space.

Uniform Plane Waves

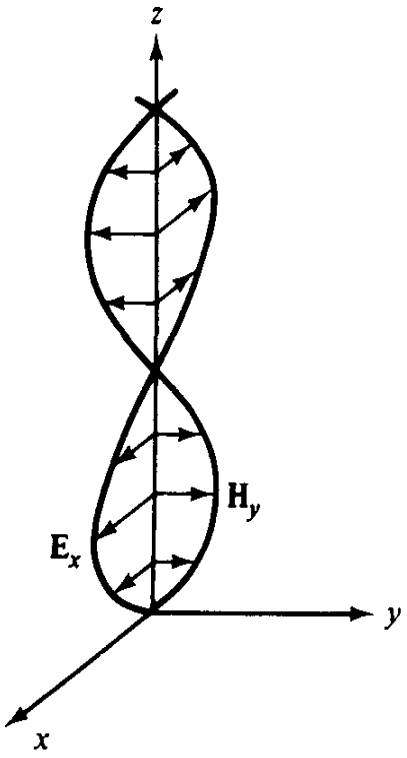


Figure 2-3-1 Uniform plane waves traveling in the z direction.

Uniform Plane Waves

Figure 2-3-1 shows uniform electric and magnetic plane waves in rectangular coordinates. In general, for a uniform plane wave propagating in a lossless dielectric medium ($\sigma = 0$), the characteristics of wave propagation would become:

$$\alpha = 0 \tag{2-3-8a}$$

$$\beta = \omega\sqrt{\mu_0\epsilon} \tag{2-3-8b}$$

$$\eta = \sqrt{\frac{\mu_0}{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon_r}} \tag{2-3-8c}$$

$$v_p = \frac{1}{\sqrt{\mu_0\epsilon}} = \frac{c}{\sqrt{\epsilon_r}} \tag{2-3-8d}$$

THANK YOU